Python Activity-Based Learning
31. Multiple-Choice Answer Evaluation

Simple student-style assignment | Easy English | Step-by-step solution

1. Problem Statement

A teacher has an answer key and a student's answers. Write a program to compare them and count correct, incorrect, and unanswered questions.

2. Objectives

• Store the answer key and student answers in lists.

• Compare answers one by one.

• Count correct, incorrect, and unanswered responses.

3. Assumptions

• There are five questions.

• A dash (-) means the student did not answer.

• Each correct answer gets one mark; there is no negative marking.

4. Algorithm (Step by Step)

1. Start.

2. Store the answer key in a list.

3. Read five student answers.

4. Set correct, incorrect, and unanswered counts to zero.

5. Compare each student answer with the matching key answer.

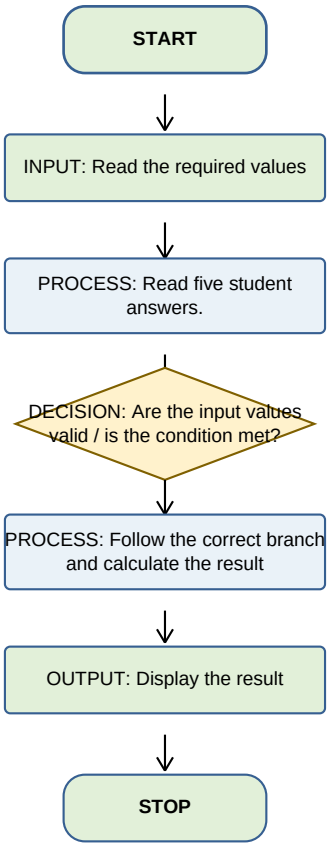
6. Update the correct, incorrect, or unanswered count.

7. Display all three counts and the final score.

8. Stop.

5. Flowchart

Follow the arrows from START to STOP. The diamond shows a decision point.



6. Pseudocode

```
START
key = [A, B, C, A, D]
INPUT five answers into answers
SET correct = 0, incorrect = 0, blank = 0
FOR i from 0 to 4
    IF answers[i] = '-' THEN blank = blank + 1
    ELSE IF answers[i] = key[i] THEN correct = correct + 1
    ELSE incorrect = incorrect + 1
PRINT counts and score
STOP
```

7. Python Source Code

```
key = ["A", "B", "C", "A", "D"]
answers = []

for i in range(5):
    answer = input("Enter answer for question " + str(i + 1) + ": ").upper()
    answers.append(answer)

correct = 0
incorrect = 0
unanswered = 0

for i in range(5):
    if answers[i] == "-":
        unanswered += 1
    elif answers[i] == key[i]:
        correct += 1
    else:
        incorrect += 1

print("Correct:", correct)
print("Incorrect:", incorrect)
print("Unanswered:", unanswered)
print("Total marks:", correct)
```

8. Test Cases

Input / Test	Expected result	What it checks
A, B, C, A, D	5 correct, 0 incorrect, 0 unanswered	All answers correct
A, -, C, X, D	3 correct, 1 incorrect, 1 unanswered	Mixed answers
-, -, -, -, -	0 correct, 0 incorrect, 5 unanswered	All blank

9. Sample Output

```
Enter answer for question 1: A
Enter answer for question 2: B
Enter answer for question 3: C
Enter answer for question 4: X
Enter answer for question 5: D
Correct: 4
Incorrect: 1
Unanswered: 0
Total marks: 4
```

10. Conclusion

This program compares two lists and counts the student's correct, incorrect, and unanswered questions.